

Application: WHSDSC26-5637003180

Geunjung Yi - yigj@csia.hs.kr

2026 Wharton High School Data Science Competition, presented by Google Gemini

Summary

ID: WHSDSC26-5637003180

Status: No action needed: Phase 1 submission is in review

Last submitted: Jan 7 2026 05:22 PM (KST)

Step 1: Submit your team's information

Completed - Jan 7 2026

Registration

Wharton High School Data Science Competition

Registration Instructions

The [Wharton Sports Analytics and Business Initiative](#) (WSABI) and [Wharton Global Youth Program](#) (WGYP) are excited to host the third Wharton High School Data Science Competition. It's a team-based opportunity for high school students to build their skills in data science through the lens of sports.

Registration Policy

- The competition is open to current high school students no younger than 14 and no older than 18 at the start of the competition.
- Each team must have a dedicated advisor who is a verified employee of the school.
- Student teams must be comprised of members from the same high school (and be from the same branch if the high school has various branches).
- Each team must have at least three and at most five students. Students can only compete on one team.
- Each team must have a Student Team Leader and this individual must be age 16 or older.
- Participating advisors are free to use this in their classroom as a learning tool. See the [Rules](#) page for detailed information.
- Each team must have completed the registration process in full (detailed below) and meet the criteria noted above to receive data and participate.

For a detailed walkthrough of the registration process, [click here](#) and watch the included video tutorial.

The registration process contains 2 steps. Step 1 must be completed by the team's Advisor. Step 2 will be completed by the Advisor and Student Team Leader.

1. In **Step 1** (where you are now), the team's Advisor will submit this application to register their team and secure their spot in the competition.

- This application will ask for information pertaining to the team's Advisor, the affiliated school, the Principal of the affiliated school, and the student team members.
- You will be asked to note which student will be serving as the Team Leader.
- The Advisor will also pick a name for the team. Please see the [Rules](#) for guidelines on team composition and team names. The team's application will need to be approved by the Wharton staff before proceeding.

2. In **Step 2**, the Advisor will add the Student Team Leader as a 'Collaborator' to the application.

- Instructions for this step will be provided after Step 1 has been completed.
- After they have been added to the application, the Student Team Leader will need to accept the invitation to collaborate.

- After they have accepted the invitation, the Student Team Leader must mark Step 2 as complete. This will automatically submit your application for review by the Wharton Team.

The team's application will need to be approved by the Wharton staff before proceeding. After your application is approved, the team will be ready for competition.

Registration steps must be completed before **January 28** to be eligible to compete. The first task (**Phase 1**) will be available on February 2.

IMPORTANT: The Student Team Leader will be responsible for accessing competition related documents and submitting their team's work through SurveyMonkey Apply. If you can't find a student to match the age criteria, please [click here](#) to notify program organizers. In order to view the competition prompt and submit work, the Student Team Leader needs to be added to this application as a 'Collaborator'.

Please feel free to [contact us](#) with any questions or concerns.

High School

High School Name	CheongShim International Academy
High School Website	www.csia.hs.kr
High School Country	South Korea
High School Postal Code	12461

Team Advisor (the designated school employee registering on behalf of the team)

Please use your school email address, if you have one, as your primary email address. This should be the same email address you used to create your SurveyMonkey Apply account. Please provide a personal email address, if you have one, as your backup email address.

First Name	GEUNJUNG
Last Name	YI
Email Address	yigj@csia.hs.kr
Backup Email Address	yigj@csia.hs.kr

Principal/Head of High School

First Name	Sung Eun
Last Name	Jun
Email	jse0001@csia.hs.kr

WSABI Communications

By providing your email address, you are agreeing to receive competition-related communications from Wharton Sports Analytics & Business Initiative (WSABI).

Responses Selected:

I agree to receive competition-related communications from WSABI

WGYP Communications

Please indicate whether you consent to also receive communications from the Wharton Global Youth Program (WGYP) in order to learn more about their programs.

I agree to receive communications from WGYP

Privacy Policy

The University of Pennsylvania provides access to services and information online in a manner that respects and protects your privacy. Our [privacy policy](#) describes our information-collection practices and explains how we use and protect your personal information.

Team Requirements

Teams must meet all criteria below in order to be eligible to compete. If you have concerns about meeting any of the criteria, [contact us](#) as soon as possible.

- Student members must be between the ages of 14 and 18 at the start of the competition (February 2)
- Teams are required to have 3-5 student members, including a Student Team Leader.
- All team members should be students at the same high school.
- Each student can only belong to one team - if a student appears on multiple teams, each team will be subject to disqualification.

The **Student Team Leader** will be responsible for submitting the team's work for each phase of the competition and this student **must be age 16 or older** in order to comply with the terms of use of SurveyMonkey Apply.

Team Composition

Please provide the information below for the students who will participate on your team.

Please indicate the number of students on your team:

3

Student Team Leader

The Student Team Leader must be age 16 or older.

Please use their school email address, if they have one. This should be the same email address you use to invite this student as a collaborator (in Step 2 of registration).

First Name	Changhoon
Last Name	Kim
Email Address	kchanghoon6@gmail.com
Graduation Year	2028

Student Team Member #2

Please use their school email address, if they have one.

First Name	GyeongHeon
Last Name	Kim
Email Address	gachikim10@gmail.com
Graduation Year	2029

Student Team Member #3

Please use their school email address, if they have one.

First Name	Geonhui
Last Name	Moon
Email Address	iuhnoeg.noom@gmail.com
Graduation Year	2029

Student Team Member Consent

By providing the students' email addresses, you are indicating that you have received consent to share the name and contact information of team members who are under the age of 18.

Responses Selected:

I have received consent to share the name and contact information of team members under age 18

Confirmation of Student Team Leader requirements

As outlined above and in competition requirements, each team is required to have one Student Team Leader who will be added to the application as a collaborator and **must be age 16 or older** in order to comply with SurveyMonkey Apply policy.

By providing a Student Team Leader email address above, you are indicating that your team meets these requirements.

Responses Selected:

I confirm that the Student Team Leader is at least 16 years old

Team Interest

Which best describes your team's interest in this competition? *Select all that apply.*

This question is for internal use only, and will not impact acceptance to the competition.

Responses Selected:

Passion for analytics primarily

Team Name

Please provide a name for your team. This cannot be changed once submitted, so choose wisely!

CSIA Data Blades

Team Intake

Did your school participate in the 2024 or 2025 Wharton High School Data Science Competition?

No

Has your school participated in any programming from Wharton Global Youth?

Yes

How did you hear about this competition? *Select all that apply.*

Responses Selected:

Social Media Post

A Wharton website or newsletter

Verification

Please check the box below to confirm:

- (1) I am an educator currently employed by the high school listed on this application.
- (2) all student team members listed on this application are currently enrolled as students in grades 9-12 at the same high school where I am employed as an educator.
- (3) all student team members listed on this application are between the ages of 14 and 18.
- (4) I understand that the students I recruit for this team cannot participate on another team.

Please note that if it is discovered at any point that all of these criteria are not met, your team will be disqualified from the competition.

Responses Selected:

As the team advisor, I confirm the above statements are all true

Submitting your application

Thank you for completing the team information! In order to successfully submit this application

- Click 'MARK AS COMPLETE' below
- Afterward, you should see information about **Step 2**: adding your Student Team Leader as a collaborator.
- When the Student Team Leader has accepted the collaborator role and marked Step 2 as complete, your team's application will automatically be submitted to the Wharton Team for review.

Step 2: Add the Student Team Leader as a collaborator

Completed - Jan 7 2026

Student Team Leaders: *please scroll down to "Instructions for the Student Team Leader to join as a 'Collaborator'" and follow the instructions listed to finalize your application.*

The final step before your team is ready to submit your registration for approval from the Wharton team is for your Student Team Leader to join your application as a 'Collaborator'.

Instructions for the Advisor to add the Student Team Leader as a 'Collaborator':

- Click the "Add Collaborator" button below and invite the Student Team Leader you indicated in the registration form as a collaborator on your application, using their school email address, if they have one. Be sure to grant them edit/view access.
- The Student Team leader will need to accept the invitation before your team will advance. This may require making a SurveyMonkey Apply account if they do not already have one.

Instructions for the Student Team Leader to join as a 'Collaborator':

- After the Student Team Leader accepts the invitation as a collaborator, they must do the following:
 - Click 'MARK AS COMPLETE' on this task, 'Step 2: Add your Student Team Leader as a collaborator'. This will automatically submit your application for review by the Wharton team.

Your team's application to the Wharton High School Data Science competition will be automatically submitted once the Student Team Leader joins as a collaborator and clicks "Mark as Complete" on this registration step.

Notes:

- Only one collaborator is permitted per application, and the collaborator must be (a) the Student Team Leader and (b) at least 16 years of age or older. Incomplete or incorrect applications will be returned for resubmission or rejected.

Once you have successfully added a collaborator and submitted your application, the status of your application will read 'Your application is in review - no action needed'. Come back on February 2 at 9:00AM EST (UTC-05:00) when the task for Phase 1 will begin.

IMPORTANT: Any team with more than one collaborator will be contacted and asked to remove all collaborators that are not the Student Team Leader. Failure to comply within 48 hours of our request will result in disqualification from the competition.

Please feel free to [contact us](#) with any questions or concerns.

Competition Prompt & Data Delivery

Completed - Feb 3 2026

Competition Prompt & Data Delivery

Wharton High School Data Science Competition: Ice Hockey Performance Predictions

Competition Prompt



You're the analytics staff for the World Hockey League. Your group is preparing for the World Hockey League (WHL) Tournament, and the WHL commissioner has a set of tasks for you. You will explore the numbers from the current WHL season* to make predictions of what teams will likely succeed in the tournament. In this competition, your group will step into the role of sports data scientists to predict tournament matchups and dive deeper to uncover features underlying team performance.

Your challenge is to formulate your responses based only on the provided, fictional World Hockey League data set. While these data are meant to reflect real-world ice hockey, they do not represent any actual teams or leagues.

**note that all data is simulated and does not represent real-world ice hockey teams or outcomes.*

How to Succeed

Success in this competition is about showcasing your skills in three key areas:

Accuracy: How effective are your matchup predictions and rankings of team depth? How clear was your visualization of the impacts of team depth on game outcomes?

Methodology: How well-thought are your quantitative approaches to creating matchup predictions and offensive depth rankings? How clear and accurate is your visualization? Can you explain your approach clearly and justify your choices with sound reasoning?

Communication: Do your findings tell a compelling story, both visually and in your presentation?

This is more than a numbers game—it's a chance to turn data into strategy, strategy into predictions, and predictions into high-level sports analytics grounded in real-world ice hockey dynamics.

Main Competition Tasks

The competition will take part in 3 phases :

Phase 1: Main Competition

Analyze stats from games across a season of the World Hockey League to rank teams and predict which teams would win in hypothetical matchups, submitted via online platform SurveyMonkey Apply.

Phase 2: Semifinals (top Student Teams invited to participate)

Creation of a short slide deck that describes and visualizes the team's methods and findings from Phase 1. Additionally, Student Teams will explore effects and impacts of line strength disparity. Slide deck will be submitted via SurveyMonkey Apply.

Phase 3: Finals (top 5 Student Teams invited to participate)

Presentation of the Phase 2 submission material to a panel of judges during a virtual meeting.

Competition Structure

About the World Hockey League

Your group will be crunching the numbers from the most recent season of the World Hockey League. You will receive one season of data, including 32 teams and 82 games per team, yielding 1,312 total games. For each game, you will see multiple rows depicting game stats broken down by matchups of home and away teams' offensive lines (first and second) and defensive pairings (first and second). Note that there are no changes in team or line quality over the course of the season, and the regular season is representative of the playoffs.

See the educational modules for more information.

Phase 1: Main Competition

The competition heats up as you prepare to dive into the action of a WHL tournament! Your mission: as an analytics group, crunch the numbers to provide an outlook for the WHL tournament, using the provided in-season data.

Your strategy and predictions will be submitted through the online platform SurveyMonkey Apply. This is where the groundwork is laid, and the sharpest solutions will advance to the next stage.

Phase 1a: Team Performance Analysis

Before diving into Phase 1a, first create a league table outlining teams' overall standings for the season following the online educational module using your method of choice (R, Python, Google Sheets, GenAI/LLM). The league table will not be submitted on its own but is a necessary first step to creating team power rankings and predicting game outcomes for your Phase 1a submission.

Create team power rankings: Based on season data, rank the 32 teams using the underlying team performance. But this isn't just about win-loss records. Your rankings should reflect the overall strength and quality of the teams.

Predict game outcomes and win probabilities: For the first round of the World Hockey League tournament, predict the win probability for the home team for 16 matchups.

Submissions: Power ranking of all 32 teams and winning probabilities of home teams in 16 matchups submitted by Student Team Leader via SurveyMonkey Apply

Phase 1b: Line Performance Analysis

Quantify team offensive line quality disparity

Hockey teams use multiple offensive lines and defensive pairings. The first offensive line is often more productive than the secondary lines — meaning they tend to generate more scoring opportunities. Your task is to quantify how large that disparity is for each team.

First, for each offensive line on each team, form an offensive performance measure based on expected goals (xG). Consider accounting for differences in time on ice so teams or lines that play more don't automatically appear better, and defensive matchups since tougher opponents can affect performance.

Second, for each team, compare performance of the first line to the secondary line by calculating a ratio of the first line's offensive performance measure to that of the secondary line. This ratio represents the team's offensive line quality disparity. Which teams have the largest offensive line quality disparity? Rank teams from largest offensive line quality disparity to smallest.

Submission: A list of the Top 10 teams with the largest offensive line quality disparity ratios (ranked 1 through 10) submitted by Student Team Leader via SurveyMonkey Apply.

Phase 1c: Data Visualization

Communicate the impact of offensive team line quality disparity

The WHL commissioner wants to know if teams with more evenly-matched offensive lines are more likely to succeed. Your task is to provide insight into how offensive line quality disparity (calculated in 1b) may relate to team strength (calculated in 1a) in a data visualization.

Your visualization should:

- Clearly communicate your main message or finding.
- Use labeled axes, titles/subtitles, captions, and legends to make your graphic understandable without additional explanation.
- Reflect sound data practices as described in the educational modules.

Submission: single PNG file upload (max file size 5MB) submitted by Student Team Leader via SurveyMonkey Apply.

Phase 1d: Tell Your Story - Summarize Your Methodology

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story. Your task is to **craft a concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that another analytics group could replicate your methodology, showcasing the rigor and transparency of your work.

Your summary should cover these key points:

(1) Process:

- How did you clean or transform the raw data before analysis? Please describe it in around 50 words.
- Did you create any additional variables? Please describe in around 25 words.

(2) Tools and Techniques:

- What software tools did you use? [Select all that apply]
- How did you use the selected software, and for what purposes? Please describe in around 50 words.
- What statistical methods did you employ? Please describe in around 100 words.

(3) Your Predictions:

- 1a: How did you determine power rankings and matchup probabilities? Please describe in around 50 words.
- 1b: How did you approach summarizing offensive line quality disparity? Please describe in around 50 words.
- 1c: What data visualization choices did you make to clarify the numbers and emphasize takeaways? Please describe in around 50 words.

(4) Your Insights:

- How did you assess your model performance (i.e., how well any models explain the data)?
- Did you use Generative AI tools (ChatGPT, Gemini, etc.)? If so, explain how– describe each stage where AI tools came into the process and how your group approached using them. Please describe in around 50 words.

Data Roadmap

To develop accurate rankings and predictions, your team will work with multiple data sets containing game statistics, team performance metrics, and matchup details. By analyzing these data sets, your team can identify trends, evaluate team strength, and explore ice hockey performance. Individual files are linked below:

Primary data set, one season of games:

Each row is the game's summary for a team - line and line pairings with descriptors of the game outcome. Each game has approximately 18 rows for the different pairings. Full season n=1,312 games.

- [Game data set \(whl 2025\)](#), primary data set, with game level line summary data
- [WHSDSC 2026 DataDictionary](#) for all fields in the Game box scores file
- [WHSDSC 2026 Glossary](#): Glossary of terms
- [WHSDSC Rnd1 matchups](#): Round 1 game matchups to predict in Phase 1a

We encourage you to download and save the provided data sets before submitting this form. However, you will also receive a PDF copy of the competition prompt, which contains important competition details, including data set descriptions, links, and key guidelines.

Additionally, all competition materials, including the **data files and project workbook**, will remain accessible throughout the competition for reference.

Your **Phase 1 submission is due on March 2nd, 2026**, 9:00am EST, so your team has a full month to explore the data, refine your analysis, and develop a strong methodology!

Use the links below to download all of the files mentioned above. **Files are provided in two ways:**

- Box [WHSDSC 2026 Hockey](#) folder (full link URL: <https://upenn.box.com/v/wsabi-hsdsc-2026>)
- Google Drive [WHSDSC 2026 CompetitionPackage](#) folder (full link URL: <https://drive.google.com/drive/folders/1WMQFKMW7yEixEfZCtTMTtS2K80IyGnRM?usp=sharing>)

Educational Modules

This competition isn't about how much you know about ice hockey or stats about your favorite team. However, we've created this video library so you can learn more about what terms mean and how they may relate to the competition. Our video library is a tool for you to reference throughout the competition.

Click the [Video Library](#) to access a playlist of all videos. A link to these videos will also be included in the .pdf workbook of the competition prompt that will be sent to you after you complete reviewing the data and competition prompt on the next page.

Educational Modules

- Educational video from Google Gemini
- Overview of WHL & game data
- Team organization
- Hockey stats: shots, xG & goals
- League Table: Overview
- League Table: Google Sheets
- League Table: R
- League Table: Python
- League Table: GenAI/LLM
- Probability: logistic model for probabilities
- Confounding
- Data visualization

Reminders

We recommend downloading and saving the provided data sets before submitting this form. However, you can access your submitted forms on the SurveyMonkey Apply platform. You will also be sent a .pdf workbook of the competition prompt and data that you can use for group work with your team.

Verification

Please check the box below to confirm that your team has read the case prompt and accessed all data files.

Responses Selected:

As the team's Student Team Leader, I confirm

Next steps

When you are ready to proceed, select 'Mark as complete' below. A PDF copy of the prompt will also be emailed to you. Please feel free to email the PDF to your fellow teammates.

You can always review the competition prompt provided here from your account dashboard.

Your Phase 1 submission is due on March 2nd, 2026, at 9:00AM ET, so your team has a month to explore the data and develop your prediction methodology!

Phase 1: Ice Hockey Performance Predictions

Completed - Feb 27 2026

Phase 1: Ice Hockey Performance Predictions

Phase 1: Ice Hockey Performance Predictions

Phase 1: Ice Hockey Performance Predictions

You're the analytics staff for the World Hockey League. Your group is preparing for the World Hockey League (WHL) Tournament, and the WHL commissioner has a set of tasks for you. You will explore the numbers from the current WHL season to make predictions of what teams will likely succeed in the tournament. In this competition, your group will step into the role of sports data scientists to predict tournament matchups and dive deeper to uncover features underlying team performance.

Your challenge is to formulate your responses based only on the provided, fictional World Hockey League data set. While these data are meant to reflect real-world ice hockey, they do not represent any actual teams or leagues.

Your analytics staff's challenge is to:

1a: Create a power ranking of WHL teams based on a season of performance data and predict the winners of tournament match-ups.

1b: Rank the top 10 WHL teams on offensive line quality disparity.

1c: Communicate the impact of offensive team line quality disparity using a data visualization.

1d: Tell your story and summarize your methodology. Craft a concise but comprehensive explanation of your approach.

You and your team have been hard at work analyzing and using the data to come up with the best answers. **Now it's time to share your predictions!**

Phase 1a: Team Performance Analysis

The competition heats up as you prepare to dive into the action of a WHL tournament! Your mission: as an analytics group, crunch the numbers to provide an outlook for the WHL tournament, using the provided in-season data.

Your strategy and predictions will now be submitted through SurveyMonkey Apply. This is where the groundwork is laid, and the sharpest solutions will advance to the next stage.

Before diving into Phase 1a, first create a league table outlining teams' overall standings for the season following the online educational module using your method of choice (R, Python, Google Sheets, GenAI/LLM). The league table will not be submitted on its own but is a necessary first step to creating team power rankings and predicting game outcomes for your Phase 1a submission.

Create team power rankings

Based on season data, rank the 32 teams using the underlying team performance. But this isn't just about win-loss records. Your rankings should reflect the overall strength and quality of the teams.

Predict game outcomes and win probabilities

For the first round of the World Hockey League tournament, predict the win probability for the home team for 16 matchups.

Phase 1a: continued

Create team power rankings

Based on your analyses, select the teams you predict will rank in each spot of the WHL tournament. **Rank the 32 teams, with 1st place being the "best" team.** The team ranking should reflect the overall quality of the team. Be sure to **double-check that the 32 teams are the ones you wish to rank and that there are no duplicate teams!** Teams must be listed in their correct ranking order.

1. Select the team you predict will rank #1 (first place) in the WHL tournament.

Brazil

2. Select the team you predict will rank #2 (second place) in the WHL tournament.

Peru

3. Select the team you predict will rank #3 (third place) in the WHL tournament.

Netherlands

4. Select the team you predict will rank #4 (fourth place) in the WHL tournament.

China

5. Select the team you predict will rank #5 (fifth place) in the WHL Tournament.

Pakistan

6. Select the team you predict will rank #6 (sixth place) in the WHL tournament.

Panama

7. Select the team you predict will rank #7 (seventh place) in the WHL tournament.

Thailand

8. Select the team you predict will rank #8 (eighth place) in the WHL tournament.

United Kingdom

9. Select the team you predict will rank #9 (ninth place) in the WHL tournament.

Iceland

10. Select the team you predict will rank #10 (tenth place) in the WHL tournament.

India

11. Select the team you predict will rank #11 (eleventh place) in the WHL tournament.

Ethiopia

12. Select the team you predict will rank #12 (twelfth place) in the WHL tournament.

Indonesia

13. Select the team you predict will rank #13 (thirteenth place) in the WHL tournament.

Philippines

14. Select the team you predict will rank #14 (fourteenth place) in the WHL tournament.

Serbia

15. Select the team you predict will rank #15 (fifteenth place) in the WHL tournament.

Guatemala

16. Select the team you predict will rank #16 (sixteenth place) in the WHL tournament.

New Zealand

17. Select the team you predict will rank #17 (seventeenth place) in the WHL tournament.

South Korea

18. Select the team you predict will rank #18 (eighteenth place) in the WHL tournament.

Saudi Arabia

19. Select the team you predict will rank #19 (nineteenth place) in the WHL tournament.

Singapore

20. Select the team you predict will rank #20 (twentieth place) in the WHL tournament.

Mexico

21. Select the team you predict will rank #21 (twenty-first place) in the WHL tournament.

Germany

22. Select the team you predict will rank #22 (twenty-second place) in the WHL tournament.

Morocco

23. Select the team you predict will rank #23 (twenty-third place) in the WHL tournament.

France

24. Select the team you predict will rank #24 (twenty-fourth place) in the WHL tournament.

Switzerland

25. Select the team you predict will rank #25 (twenty-fifth place) in the WHL tournament.

Vietnam

26. Select the team you predict will rank #26 (twenty-sixth place) in the WHL tournament.

Oman

27. Select the team you predict will rank #27 (twenty-seventh place) in the WHL tournament.

Canada

28. Select the team you predict will rank #28 (twenty-eighth place) in the WHL tournament.

United States of America

29. Select the team you predict will rank #29 (twenty-ninth place) in the WHL tournament.

United Arab Emirates

30. Select the team you predict will rank #30 (thirtieth place) in the WHL tournament.

Kazakhstan

31. Select the team you predict will rank #31 (thirty-first place) in the WHL tournament.

Rwanda

32. Select the team you predict will rank #32 (thirty-second place) in the WHL tournament.

Mongolia

Phase 1a: continued

Predict game outcomes and win probabilities

Predict the win probabilities for the home team (listed first) for 16 first round WHL tournament matchups. This is your chance to flex your data-driven strategy skills! Your predictions should focus on the stats and strategies that truly matter, not just the team's standing.

When you submit your predictions, your submission for each game should be a numeric probability between 0 and 1. Think of it as assigning the home team a “chance to win” based on your analysis.

For each game, estimate and input the probability that the home team (listed first) wins. Each of your answers should be a numeric value between 0 and 1. For example, if you think a team has a 67% chance of winning, write .67 as your answer.

Game 1: Brazil - Kazakhstan

Estimate the probability that **Brazil** wins:

.80

Game 2: Netherlands - Mongolia

Estimate the probability that **Netherlands** wins:

.78

Game 3: Peru - Rwanda

Estimate the probability that **Peru** wins:

.80

Game 4: Thailand - Oman

Estimate the probability that **Thailand** wins:

.73

Game 5: Pakistan - Germany

Estimate the probability that **Pakistan** wins:

.70

Game 6: India - United States

Estimate the probability that **India** wins:

.70

Game 7: Panama - Switzerland

Estimate the probability that **Panama** wins:

.68

Game 8: Iceland - Canada

Estimate the probability that the **Iceland** wins:

.70

Game 9: China - France

Estimate the probability that the **China** wins:

.70

Game 10: Philippines - Morocco

Estimate the probability that **Philippines** wins:

.62

Game 11: Ethiopia - Saudi Arabia

Estimate the probability that **Ethiopia** wins:

.62

Game 12: Singapore - New Zealand

Estimate the probability that the **Singapore** wins:

.56

Game 13: Guatemala - South Korea

Estimate the probability that **Guatemala** wins:

.59

Game 14: United Kingdom - Mexico

Estimate the probability that **United Kingdom** wins:

.66

Game 15: Vietnam - Serbia

Estimate the probability that **Vietnam** wins:

.50

Game 16: Indonesia - United Arab Emirates

Estimate the probability that **Indonesia** wins:

.68

Phase 1b: Line Performance Analysis

Quantify team offensive line quality disparity

Hockey teams use multiple offensive lines and defensive pairings. The first offensive line is often more productive than the secondary lines — meaning they tend to generate more scoring opportunities. Your task is to quantify how large that disparity is for each team.

First, for each offensive line on each team, form an *offensive performance measure* based on expected goals (xG). Consider accounting for differences in time on ice so teams or lines that play more don't automatically appear better, and defensive matchups since tougher opponents can affect performance.

Second, for each team, compare performance of the first line to the secondary line by calculating a ratio of the first line's offensive performance measure to that of the secondary line. This ratio represents the team's *offensive line quality disparity*. Which teams have the largest offensive line quality disparity?

Rank the top 10 NHL teams on offensive line quality disparity from largest to smallest, where 1 = largest offensive line quality ratio.

1. Select the WHL team with the #1 (largest) offensive line quality disparity ratio.

United States of America

2. Select the WHL team with the #2 (second-largest) offensive line quality disparity ratio.

Guatemala

3. Select the WHL team with the #3 (third-largest) offensive line quality disparity ratio.

Saudi Arabia

4. Select the WHL team with the #4 (fourth-largest) offensive line quality disparity ratio.

United Arab Emirates

5. Select the WHL team with the #5 (fifth-largest) offensive line quality disparity ratio.

France

6. Select the WHL team with the #6 (sixth-largest) offensive line quality disparity ratio.

Iceland

7. Select the WHL team with the #7 (seventh-largest) offensive line quality disparity ratio.

Singapore

8. Select the WHL team with the #8 (eighth-largest) offensive line quality disparity ratio.

New Zealand

9. Select the WHL team with the #9 (ninth-largest) offensive line quality disparity ratio.

Peru

10. Select the WHL team with the #10 (tenth-largest) offensive line quality disparity ratio.

Panama

Phase 1c: Data Visualization

Communicate the impact of offensive team line quality disparity

The WHL commissioner wants to know if teams with more evenly-matched offensive lines are more likely to succeed. Your task is to provide insight into how offensive line quality disparity (calculated in 1b) may relate to team strength (calculated in 1a) in a data visualization.

Your visualization should:

- Clearly communicate your main message or finding.
- Use labeled axes, titles/subtitles, captions, and legends to make your graphic understandable without additional explanation.
- Reflect sound data practices as described in the educational modules.

Please upload your data visualization in a single PNG file upload (max file size 5MB) titled with your team name without spaces (e.g., DataScienceTeam.png).

[WHSDSC_2026.png](#)

Filename: WHSDSC_2026.png.png **Size:** 92.1 kB

Phase 1d: Tell Your Story - Summarize Your Methodology

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Process

How did you clean or transform the raw data before analysis? Please describe it in around 50 words.

We ingested the season dataset, standardized team identifiers, and verified integrity by confirming each game_id had exactly one home and one away record. We validated the schedule structure (32 teams, 82 games per team, 1312 total). We then aggregated to game level and converted xG and TOI into per-60 rates for consistent line comparisons.

Did you create any additional variables? Please describe in about 25 words.

We derived xG/60 for each offensive line, a first-to-second line disparity ratio, home-ice and overtime indicators, and model-based team strength scores.

Phase 1d: continued

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Tools & Techniques

What software tools did you use? Select all that apply.

Responses Selected:

Excel
Google Sheets
Python
Other, please specify...: vs code

How did you use the selected software, and for what purposes? Please describe in around 50 words.

Python handled data cleaning, joins, and modeling, producing reproducible CSV and PNG outputs. Excel and Google Sheets were used to spot-check tables, verify team/game counts, and review rankings and probabilities. VS Code served as the development environment for running scripts, tracking changes, and organizing the pipeline.

What statistical methods did you employ? Please describe in around 100 words.

We estimated team strength with a Poisson generalized linear model (log link), using team indicators for attack and defensive-allowance effects plus a home-ice term. Expected goals for each Round 1 matchup were derived from fitted parameters, then converted to home-win probabilities using a Skellam goal-differential model. Offensive line disparity was defined as $\text{first_off xG}/60$ divided by $\text{second_off xG}/60$ after TOI normalization. Uncertainty was quantified by bootstrapping games to form 95% confidence intervals, and association checks used Spearman rank correlation. We evaluated probability quality with Brier score and log loss on season outcomes and compared baseline, adjusted, and regression disparity estimators to confirm rank stability.

Phase 1d: continued

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Predictions

1a: How did you determine power rankings and matchup win probabilities? Please describe in around 50 words.

We produced power rankings from a model-based strength score rather than raw win-loss. A Poisson GLM estimates team attack and defensive-allowance effects and yields expected goals for each matchup. Using these rates, we computed Round 1 home-win probabilities from the Skellam goal-differential distribution, producing matchup-specific probabilities aligned with the scoring model.

1b: How did you approach summarizing offensive line quality disparity? Please describe in around 50 words.

We quantified line quality using xG/60 for each offensive line and computed a disparity ratio ($\text{first_off}/\text{second_off}$). To reduce confounding from usage and opponent effects, we used regression-based rate estimates and ranked teams by the resulting disparity ratios. We also quantified uncertainty via bootstrap confidence intervals to distinguish robust disparities from estimates with substantial variability.

1c: What data visualization choices did you make to clarify the numbers and emphasize takeaways? Please describe in around 50 words.

We used a single scatter plot to summarize the relationship between team strength and offensive-line disparity. To preserve readability, we labeled only a subset of teams while plotting all teams. Axes are explicitly defined, and uncertainty is emphasized by indicating whether each team's 95% CI crosses the no-disparity threshold of 1.0.

Phase 1d: continued

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Your Insights

How did you assess your model performance? Please describe in around 50 words.

We evaluated predictive performance using log loss and Brier score on game outcomes, with 5-fold cross-validation to assess calibration. We checked model stability with standard GLM diagnostics and enforced pipeline QA checks: expected row counts, successful joins across derived tables, valid CI bounds, and output-file size limits. These checks support correctness and reproducibility.

Did you use Generative AI tools (ChatGPT, Gemini, etc.)? If so, explain how— describe each stage where AI tools came into the process and how your group approached using them. Please describe in around 50 words.

We used a writing assistant only to refine wording for clarity. All rankings, probabilities, statistical estimates, and figures were generated by our scripts from the provided dataset. We validated outputs by rerunning the full pipeline and confirming the same integrity checks passed.

Verify Your Team Information

Before submitting your answers for Phase 1, please reconfirm your team member details. We understand that teams may have changed since the initial registration period, so this is your opportunity to ensure that the correct names are listed.

These will be the official team details used for the remainder of the competition and for generating certificates of participation—so please double-check that everything is entered correctly!

Team Composition

Please select the number of students on your team.

3

Student Team Leader

Student Team Leader: please enter your school email address here, if you have one. This should be the same email address that you used to sign into SurveyMonkey Apply to work on this application as a collaborator.

Student Team Leaders cannot be changed. Please email data-comp@wharton.upenn.edu with any questions or concerns.

First Name	Changhoon
Last Name	Kim
Email Address	kchanghoon6@gmail.com
Graduation Year	2028

Student Team Member #2

Please use their school email address, if they have one.

First Name	GyeongHeon
Last Name	Kim
Email Address	gachikim10@gmail.com
Graduation Year	2029

Student Team Member #3

Please use their school email address, if they have one.

First Name	Geonhui
Last Name	Moon
Email Address	iuhnoeg.noong@gmail.com
Graduation Year	2029

Last Step - Feedback Survey

In the final part of Phase 1, we'd like to know more about how your team utilized the resources provided by our team for the competition and how your team approaches it. Your responses help us make changes to improve the competition for future teams. Please note that **your answers to these questions will *not* affect your standing in the competition.**

Did you use any of the educational materials provided for this competition?

No

To what degree were the educational modules helpful to your team?

Please use the following scale where **0** = *not at all helpful* and **10** = *extremely helpful*

If you did not watch the educational modules, please select **NA**.

NA

Around how many hours a *week* did your team spend working on this project?

7 - 10 hours

Around how many *TOTAL* hours did your team spend working on this project?

30 - 40 hours

Is there anything else you'd like to share about your team's experience with Phase 1—whether about the data, the process, your approach, or something unexpected your team encountered?

The dataset was clean and well-structured, which let us focus on modeling rather than wrangling. A pre-supplied data dictionary would have helped clarify column definitions early on. We enjoyed the open-ended format—it encouraged creative approaches while keeping the scope manageable.

Phase 1: Submit Your Answers

After submitting this form, you will have officially completed Phase 1 of the Wharton High School Data Science Competition! Please ensure all your answers are entered correctly, as you will not be able to re-submit once your responses are submitted.

Competition Semifinalists will be announced **on Monday, March 9th**. All teams will receive a message indicating whether they advanced to Phase 2 or did not advance.

Please check the box to confirm:

- (1) the submitted Phase 1 answers are final,
- (2) the submitted student team information is accurate and final, and
- (3) the student team members listed are the individuals who completed the analysis and work for the Wharton High School Data Science Competition.

Responses Selected:

As the Student Team Leader, I confirm

Please click "MARK AS COMPLETE" below to submit your Phase 1 answers. You will not be able to edit your responses after this step.